UPS HAT (E) Register

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Introduction

- It is easy to operate and use as the UPS HAT (E) can read module information through I2C communication. The module slave address: 0x2D.

Register Table

- ID Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x00	R	Fixed value: 0x0a							

- Automatically Rebooting Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x01	R/W	Fixed value: 0x0b, write "0x55" to power off and enable the incoming power function							

- Charging Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x02	R	1: charging 0: non-charging	1: fast charging 0: no fast charging	1: VBUS is powered 0: VBUS is not powered	Fixed: 00		000: standby, 001: trickle charge, 010: constant current charge; 011: constant voltage charge; 100: Charging pending, 101: Full state, 110: Charge timeout		

- Communication Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x03	R	Fixed: 000000						1: BQ4050 communication is normal; 0: BQ4050 communications error	1: IP2368 communication is normal; 0: IP2368 communication error

- Type-C VBUS Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x10	R	VBUS voltage data low 8bit(mV)							
0x11	R	VBUS voltage data high 8bit(mV)							

- Type-C VBUS Current Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x12	R	VBUS current data low 8bit(mA)							
0x13	R	VBUS current data high 8bit(mA)							

- Type-C VBUS Power Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x14	R	VBUS power data low 8bit(mW)							
0x15	R	VBUS power data high 8bit(mW)							

- Battery Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x20	R	Battery total voltage data low 8bit(mV)							
0x21	R	Battery total voltage data high 8bit(mV)							

- Battery Current Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x22	R	Battery current data low 8bit(mA)							
0x23	R	Battery current data high 8bit(mA) Signed sixteen bits, positive indicates charging current, negative indicates output current							

- Battery Percent Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x24	R	Battery percent data low 8bit(%)							
0x25	R	Battery percent data high 8bit(%)							

- Battery Remaining Capacity Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x26	R	the low 8bit(mAh) of battery remaining capacity data							
0x27	R	the high 8bit(mAh) of battery remaining capacity data							

- Battery Remaining Discharge Time Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x28	R	the low 8bit(min) of the battery remaining discharge time data							
0x29	R	the high 8bit(min) of the battery remaining discharge time data							

- Battery Remaining Charging Time Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x2a	R	The low 8bit(min) of the battery remaining charging time data							
0x2b	R	The high 8bit(max) of the battery remaining charging time data							

- Cell 1 Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x30	R	The low 8bit(mV) of the cell 1 voltage data							
0x31	R	The high 8bit(mV) of the cell 1 voltage data							

- Cell 2 Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x32	R	The low 8bit(mV) of the cell 2 voltage data							
0x33	R	The high 8bit(mV) of the cell 2 voltage data							

- Cell 3 Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x34	R	The low 8bit(mV) of the cell 3 voltage data							
0x35	R	The high 8bit(mV) of the cell 3 voltage data							

- Cell 4 Voltage Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x36	R	The low 8bit(mV) of the cell 4 voltage data							
0x37	R	The high 8bit(mV) of the cell 4 voltage data							

Register Table V2 (Addition)

- Control Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x40	R/W	Reserved						1: Enable Watchdog 0: Disable Watchdog (default)	1: auto-restart upon power incoming (default) 0: Disable

- Device addrees Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x41	R/W	I2c device address: 0x08 ~ 0x77; Default: 0x2d							

- Watchdog timeout Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x42	R/W	Timeout: 0 ~ 255; unit: s ; Default: 0xFF							

- Watchdog delay start-up Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x43	R/W	Dealy time: 0 ~ 255; unit: s ; Default: 0xFF							

- SW Register

Register	R/W	BIT7	BIT6	BIT5	BIT4	BIT3	BIT2	BIT1	BIT0
0x50	R	Software revision. For examples: 0x14 = 20 (DEC) = V2.0							

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